Mahadevan Reji

+91-9188396609 | mahadevan574@gmail.com | linkedin.com/in/mahadevanreji | github.com/kichu12348

Education

College of Engineering Chengannur — Chengannur, Kerala
Bachelor of Technology in Computer Science and Engineering — *Sept. 2023 – Apr. 2027*

Experience

Web and Mobile App Developer Intern — June 2025 – July 2025
ZenturioTech — *Technopark, Thiruvananthapuram*

- Engineered on-device inference capabilities within a React Native application by integrating machine learning models via TensorFlow.js, reducing network dependency and latency
- Executed the end-to-end development of "Kids Art Studio," an educational app featuring automated drawing analysis and real-time feedback loops

Projects

NotChat | *React Native, TypeScript, Cloudflare* — July 2026 – Present

- Architected a full-stack E2EE chat application with real-time WebSocket messaging on a serverless edge architecture
- Implemented client-side encryption via X25519 key exchange and AES-256-GCM, with ephemeral 24-hour message storage for zero-persistence privacy

Bruno | *TypeScript, Radix Tree, AOT* — June 2026 – July 2026

- Designed an AOT-compiled radix tree router that translates route definitions into optimized if-else chains, eliminating runtime lookup overhead
- Achieved peak throughput of 300,000 req/s, tuned for latency-critical edge runtimes

NanoDB | *Go, B+ Trees, Vector Search* — Jan. 2026 – Feb. 2026

- Built a custom storage engine from scratch using a slotted page architecture for space-efficient on-disk data storage
- Implemented B+ Tree indexing alongside native vector search, enabling efficient similarity-based queries over structured data

BunkMate | *React Native, Expo* — June 2025 – Present

- Engineered a local-first architecture serving 180+ users with centralized backend synchronization
- Owned the full product lifecycle from ideation to deployment, iterating on features based on direct user feedback

SnapBook | *React Native, Node.js, WebSockets* — June 2025 – July 2025

- Built real-time synchronization using WebSockets and an event-driven architecture
- Enabled seamless multi-user collaboration across the platform

Technical Skills

Languages: TypeScript, JavaScript, Go, Rust, Python, SQL
Mobile & Frontend: React Native, Expo, React
Backend & Edge: Node.js, Bun, Fastify, Hono, Axum, FastAPI
Developer Tools: Git, Docker, Google Cloud Platform, VS Code
Libraries: pandas, NumPy
Core Engineering: B+ Trees, Storage Engines, Slotted Page Architecture, Vector Search, AOT Compilation